

M003 Week 3 — English Worksheet

Topic: Pixel Art on the Wall (x, y) · **Course:** 3D Coordinates · **Time:** about 45 minutes

This week you draw **pictures with blocks** on the wall. A picture is made of **rows**, and a row is made of colors in order. A **nested loop** — a loop inside a loop — draws the picture row by row: the inner loop grows **x** across, the outer loop grows **y** up.

1 · Predict 🧠

Read each set of steps. Before you imagine it happening, write what you think you will see.

```
set y to 0
repeat 2 times:
  set x to 0
  repeat 3 times:
    place red block at (x, y)
    add 1 to x
  add 1 to y
```

How many red blocks are in each row? How many rows? How many blocks in total?


```
set x to 0
repeat 2 times:
  place red block at (x, 0)
  add 1 to x
place blue block at (x, 0)
```

This draws one row. Say the colors in order — what does the bottom row look like?


```
set y to 0
repeat 3 times:
  place red block at (0, y)
  place yellow block at (1, y)
  place red block at (2, y)
  add 1 to y
```

Every row is red, yellow, red. What does the whole picture look like? Which color makes a stripe straight up the middle?

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — This should draw **3 rows** of blue blocks, each row one step higher. Right now every row lands in the **same place**.

```
set y to 0
repeat 3 times:
  set x to 0
  repeat 4 times:
    place blue block at (x, y)
    add 1 to x
```

Hint: after each row, the picture needs to step up to the next y.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The bottom row should look like **red, red, blue** — red at (0, 0) and (1, 0), blue at (2, 0). Right now the colors come out in the wrong order.

```
place blue block at (0, 0)
place red block at (1, 0)
place red block at (2, 0)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The picture should be **5 blocks wide** and 3 rows tall. Right now it comes out squashed and thin.

```
set y to 0
repeat 3 times:
  set x to 0
  repeat 2 times:
    place green block at (x, y)
    add 1 to x
  add 1 to y
```

Hint: the inner loop controls how wide each row is.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now switch to your homework world. Draw a **pixel art picture** on the wall — a heart, a face, or the first letter of your name — using rows of colored blocks. When you finish, come back here.

Send a photo or video of your picture, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I planned my picture by ...

My picture is ... rows tall and ... blocks wide.

The colors in the bottom row are ...

I used a nested loop when ...

One tricky moment was when ...

If I had more time, I would ...

4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while you show your pixel art on the wall. Talk through it like you are teaching someone who has never seen it. Try to use these words: **row**, **nested loop**, **inner**, **outer**, **pattern**.

1. Show your finished picture from the front.
2. Point at the bottom row and say the colors in order.
3. Show your code and say what the **inner** loop does (across) and what the **outer** loop does (up).
4. Say in your own words why the picture is drawn row by row.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.